

Standard Deviation of a Standardized Vector

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Assume $\mathbf{x}$ is not a constant vector, so $\text{std}(\mathbf{x}) \neq 0$. Therefore, the vector $\mathbf{z}$, the *standardized* version of $\mathbf{x}$, is well defined.

$$\begin{aligned} \mathbf{z} &= \frac{\tilde{\mathbf{x}}}{\text{std}(\mathbf{x})} \\ &= \frac{\sqrt{n}}{\|\tilde{\mathbf{x}}\|} \tilde{\mathbf{x}} \quad \because \text{std}(\mathbf{x}) = \frac{\|\tilde{\mathbf{x}}\|}{\sqrt{n}} \end{aligned}$$

Since $\mathbf{z}$ is a positive scalar multiple of $\tilde{\mathbf{x}}$, and scaling preserves the average.

It is easy to see why scaling preserves the average:

$$\begin{aligned} \mathbf{x} &= (x_1, x_2, \dots, x_n) \\ \text{avg}(\mathbf{x}) &= \frac{x_1 + x_2 + \dots + x_n}{n} \\ \alpha \mathbf{x} &= (\alpha x_1, \alpha x_2, \dots, \alpha x_n) \\ \text{avg}(\alpha \mathbf{x}) &= \frac{\alpha x_1 + \alpha x_2 + \dots + \alpha x_n}{n} \\ &= \frac{\alpha (x_1 + x_2 + \dots + x_n)}{n} \\ &= \alpha \text{avg}(\mathbf{x}) \end{aligned}$$

Therefore,

$$\begin{aligned}
 \text{avg}(\mathbf{z}) &= \text{avg}\left(\frac{\sqrt{n}}{\|\tilde{\mathbf{x}}\|} \tilde{\mathbf{x}}\right) \\
 &= \left(\frac{\sqrt{n}}{\|\tilde{\mathbf{x}}\|}\right) \text{avg}(\tilde{\mathbf{x}}) \\
 &= 0 \qquad \because \text{avg}(\tilde{\mathbf{x}}) = 0
 \end{aligned}$$

Let us try to determine $\text{std}(\mathbf{z})$.

$$\begin{aligned}
 \text{std}(\mathbf{z}) &= \frac{\|\tilde{\mathbf{z}}\|}{\sqrt{n}} \\
 &= \frac{\|\mathbf{z}\|}{\sqrt{n}} \qquad \because \text{avg}(\mathbf{z}) = 0 \\
 &= \frac{1}{\sqrt{n}} \left\| \frac{\sqrt{n}}{\|\tilde{\mathbf{x}}\|} \tilde{\mathbf{x}} \right\| \\
 &= \frac{1}{\sqrt{n}} \cdot \frac{\sqrt{n}}{\|\tilde{\mathbf{x}}\|} \cdot \|\tilde{\mathbf{x}}\| \qquad \because \|c \mathbf{v}\| = |c| \|\mathbf{v}\| \text{ for } c > 0 \\
 &= 1
 \end{aligned}$$